USING DISCLOSURES IN ANALYSIS

Learning Outcome

- analyze and interpret financial statement disclosures regarding property, plant, and equipment and intangible assets

Ratios used in analyzing fixed assets include the fixed asset turnover ratio and several asset age ratios. The fixed asset turnover ratio (total revenue divided by average net fixed assets) reflects the relationship between total revenues and investment in PPE (property, plant, & equipment). The higher this ratio, the higher the amount of sales a company is able to generate with a given amount of investment in fixed assets. A higher asset turnover ratio is often interpreted as an indicator of greater efficiency.

Asset age ratios generally rely on the relationship between historical cost and depreciation. Under the revaluation model (permitted under IFRS but not US GAAP), the relationship between carrying amount, accumulated depreciation, and depreciation expense will differ when the carrying amount differs significantly from the depreciated historical cost. Therefore, the following discussion of asset age ratios applies primarily to PPE reported under the cost model.

Asset age and remaining useful life, two asset age ratios, are important indicators of a company's need to reinvest in productive capacity. The older the assets and the shorter the remaining life, the more a company may need to reinvest to maintain productive capacity. The average age of a company's asset base can be estimated as accumulated depreciation divided by depreciation expense. The average remaining life of a company's asset base can be estimated as net PPE divided by depreciation expense. These estimates simply reflect the following relationships for assets accounted for on a historical cost basis: total historical cost minus accumulated depreciation equals net PPE; and, under straight-line depreciation, total historical cost less salvage value divided by estimated useful life equals annual depreciation expense. Equivalently, total historical cost less salvage value divided by annual depreciation expense equals estimated useful life. Assuming straight-line depreciation and no salvage value (for simplicity), we have the following:

Estimated total useful life	=	Time elapsed since purchase (Age)	+	Estimated remaining life
Historical cost ÷ annual depreciation expense	=	Estimated total useful life		
Historical cost	=	Accumulated depreciation	+	Net PPE

Equivalently,

Estimated total useful life	=	Estimated age of equipment	+	Estimated remaining life
Historical cost ÷ annual depreciation expense	=	Accumulated depreciation ÷ annual depreciation expense	+	Net PPE ÷ annual depreciation expense

The application of these estimates can be illustrated by a hypothetical example of a company with a single depreciable asset. Assume the asset initially cost USD100, had an estimated useful life of 10 years, and an estimated salvage value of USD0. Each year, the company records a depreciation expense of USD10, so accumulated depreciation will equal USD10 times the number of years since the asset was acquired (when the asset is 7

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In practice, such estimates are difficult to make with great precision. Companies use depreciation methods other than the straight-line method and have numerous assets with varying useful lives and salvage values, including some assets that are fully depreciated, so this approach produces an estimate only. Moreover, fixed asset disclosures are often quite general. Consequently, these estimates may be primarily useful to identify areas for further investigation.

One further measure compares a company's current reinvestment in productive capacity. Comparing annual capital expenditures to annual depreciation expense provides an indication of whether productive capacity is being maintained. It is a very general indicator of the rate at which a company is replacing its PPE relative to the rate at which PPE is being depreciated.

EXAMPLE 6

Using Fixed Asset Disclosure to Compare Companies' Fixed Asset Turnover and Average Age of Depreciable Assets

Consider the property, plant, and equipment for the following three international telecommunications companies:

- Orange SA, which we discussed previously, has been listed on Euronext Paris (symbol ORA) and on the New York Stock Exchange (symbol ORAN) since 1997. At 31 December 2017, the French government retained 22.95 percent of the share capital.
- BCE Inc., Canada's largest communications company, provides wireless, wireline, Internet, TV, and business communications across Canada. BCE's shares are publicly traded on the Toronto Stock Exchange and on the New York Stock Exchange (TSX, NYSE: BCE).
- Verizon Communications Inc. is a US-based global provider of communications, information, and entertainment products and services to consumers, businesses, and governmental agencies. Verizon's shares are listed on the New York Stock Exchange and the NASDAQ Global Select Market (symbol VZ).

Exhibit 6:

Selected Information from the Companies' Financial Statements

	Orange	BCE Inc	Verizon
Currency, Millions of:	**Euro**	**Canadian dollars**	**US dollars**
Historical cost total PPE, end of year	97,092	69,230	246,498
Accumulated depreciation, end of year	70,427	45,197	157,930
Net PPE, end of year	26,665	24,033	88,568
Net PPE, beginning of year	25,912	22,346	84,751
Revenues	41,096	22,719	126,034
Annual depreciation expense	4,708	3,037	14,741
Capital expenditure	5,677	4,149	17,247
Land included in PPE	Not separated	Not separated	806
Accounting standards	IFRS	IFRS	US GAAP
PPE measurement	Historical cost	Historical cost	Historical cost

Currency, Millions of:	Euro	dollars	US dollars
Depreciation method	Straight-line	Straight-line	Straight-line

Sources: Companies' 2017 Annual Financial Reports."

1. Based on the data for each company in Exhibit 5, estimate the total useful life, age, and remaining useful life of PPE.

Hide Solution

Solution:

The following table presents the estimated total useful life, estimated age, and estimated remaining useful life of PPE for each of the companies.

Estimates	Orange	BCE Inc	Verizon
Estimated total useful life (years)	20.6	22.8	16.7
Estimated age (years)	15.0	14.9	10.7
Estimated remaining life (years)	5.7	7.9	6.0

The computations are demonstrated using Verizon's data ($ in millions). The estimated total useful life of PPE is total historical cost of PPE of USD246,498 divided by annual depreciation expense of USD14,741, giving 16.7 years. Estimated age and estimated remaining life are obtained by dividing accumulated depreciation of USD157,930 and net PPE of USD88,568 by the annual depreciation expense of USD14,741, giving 10.7 years and 6.0 years, respectively.

Ideally, the estimates of asset lives illustrated in this example should exclude land, which is not depreciable, when the information is available; however, both Orange and BCE report Land and Buildings as a combined amount. We will use Verizon, for which land appeared to be disclosed separately in the above table, to illustrate the estimates with adjusting for land. As an illustration of the calculations to exclude land, excluding Verizon's land would give an estimated total useful life for the non-land PPE of 16.7 years [(total cost EUR246,498 minus land cost of USD806) divided by annual depreciation expense of EUR14,741 million]. The estimate is essentially unchanged from the estimate including land because land represents such a small component of Verizon's PPE.

2. Interpret the estimates of estimated total useful life, age, and remaining life. What items might affect comparisons across the three companies?

Hide Solution

Solution:

The estimated total useful life suggests that Orange and BCE depreciate PPE over a much longer period than Verizon: 20.6 and 22.8 years for Orange and BCE, respectively, versus 16.7 years for Verizon.

The estimated age of the equipment suggests that Verizon has the newest PPE with an estimated age of 10.7 years. Additionally, the estimates suggest that around 73 percent of Orange's assets' useful lives have passed (15.0 years ÷ 20.6 years, or equivalently, EUR70,427 million ÷ EUR97,092 million). In comparison, around 65 and 64 percent of the useful lives of the PPE of BCE and Verizon, respectively, have passed.

Items that can affect comparisons across the companies include business differences, such as differences in composition of the companies' operations and differences in acquisition and divestiture activity. This result can be compared, to an extent, to the useful lives and asset mix disclosed in the

3. How does each company's 2017 depreciation expense compare to its capital expenditures for the year?

Hide Solution

Solution:

All three companies' capital expenditure exceeds its depreciation expense. Rounding to the nearest 10 percent, capital expenditure as a percentage of depreciation is 120 percent for Orange, 140 percent for BCE, and 120 percent for Verizon. All three companies are replacing PPE at a faster rate than the PPE is being depreciated, consistent with the companies' somewhat older asset base.

4. Calculate and compare fixed asset turnover for each company.

Hide Solution

Solution:

Fixed asset turnover is calculated as total revenues divided by average net PPE. Orange's fixed asset turnover is 1.6 (= 41,096/((26,665 + 25,912)/2)). BCE's fixed asset turnover is 1.0, and Verizon's fixed asset turnover is 1.5.

Orange's and Verizon's higher levels of fixed asset turnover indicate these companies, compared to BCE, are able to generate more sales per unit of investment in fixed assets.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

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{FLDS} FIELD SEARCH

The Bloomberg FLDS function allows you to view Bloomberg's data fields, including different depreciation methods for property, plant, and equipment, and impairments and revaluations for public companies.

> "FLDS <GO> is your hub for exploring Bloomberg data, providing a breadth of fields for legacy APIs, calcrts, and the Bloomberg Query Language (BQL) in one centralized location." -Bloomberg

To explore this function, type FLDS into the Bloomberg command line and press Enter or <GO>. Type a security—such as Unilever PLC (or ULVR LN Equity)—into the top left amber field, and press Enter or <GO>.

- For depreciation method, type "depreciation method" into the Enter Query amber field, press Enter or <GO>, and review the depreciation method used for the majority of fixed assets for the chosen company.
- For impairment, type "impairment" into the Enter Query amber field, press Enter or <GO>, and review the impairments for the chosen company.
- For revaluation, type "revaluation" into the Enter Query amber field, press Enter or <GO>, and review the revaluations for the chosen company.

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1)	DX829	DEPRECIATION_METHOD		Straight line method
2)	IS938	IS_EBITDA		N.A.
3)	BS030	BS_GROSS_FIX_ASSET		23,121.00
4)	FS321	MINING_GOLD_CASH_COSTS…		N.A.
5)	IS972	IS_ADJUSTED_EBITDA_AS_…		11,408.00
6)	FS921	MINING_GOLD_CASH_COSTS…		N.A.
7)	CF012	CF_OTHER_NON_CASH_ADJU…		-2,156.00
8)	FS406	MINING_COPPER_CASH_COS…		N.A.
9)	FS334	MINING_SILVER_CASH_COS…		N.A.
10)	FS934	MINING_COPPER_CASH_COS…		N.A.
11)	FS929	MINING_SILVER_CASH_COS…		N.A.
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